



SKULLBONEZ

CORE

FIRST-TIME USER MANUAL

Explore. Build a scene. Test what happens.

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Your first session

Thank you for trying the very first public release of SkullbonezCore. This guide helps you explore, build and test your own scenes.

1. Extract, then launch

Download the Windows ZIP from the [public release page](#). Extract the entire archive to a writable folder and run **Launch SkullbonezCore.cmd**. Keep the executable, DLLs and **SkullbonezData** together. You need 64-bit Windows and a DirectX 12-compatible GPU and driver. This build is unsigned; no development tools are needed.

2. Learn to look and move

Choose **Start tutorial** on first launch, or click **Tutorial** in the top bar. The first lesson teaches the camera before you enter the car. Hold **right mouse** over the scene and move the mouse to look around. Keep it held and use **W/A/S/D** to move. Release right mouse to use panels and tools again.

3. Find the controls

Press **Esc** to open the menu, then open **Editor** at the left edge. **Tools > Scene** loads levels; **Tools > Keys** shows controls and tutorial Resume/Restart. Hover for help and scroll inside panels. The top-right header shows Objects, then Scene/Frame on the next line.

Start with these controls

Control	What it does
Right mouse + move mouse	Look around the viewport. In an orthographic view, pan instead.
Right mouse + W/A/S/D	Move forward, left, backward and right. Q/E moves down/up.
Shift / wheel while looking	Temporary speed boost / adjust flight speed.
Middle-drag / idle wheel	Pan the camera / zoom.
Double-click an object	Attach the camera. For a car, press T to enter Drive.
W/A/S/D / Space in Drive	Accelerate, steer, reverse / brake. Esc detaches.
Edit scene	Enter or leave editing; a filled button means it is active.
0 (zero)	Show or hide the UI.

Good to know

The tutorial covers driving, rewinding, editing, terrain and the Solar System. **Retry section** restores the current chapter; **Skip step** moves past a lesson. Practice scenes are protected: use **New scene** to make a level you can save.



Build your first scene

Example project: a box and a ball on flat terrain. No JSON editing is required.

1. Open the Editor

Press **Esc** if you need the menu, open **Editor**, then choose **Objects**. Save any existing work before starting a new level.

2. Create a new scene

Click **New scene**. This opens a blank scene with editing enabled. Remember its name so you can find the level again in **Tools > Scene**.

3. Find the placement controls

Choose a type from the scrollable object list to enter placement automatically. **Edit scene** controls editing; **Select [Q]** returns to selecting existing objects.

4. Place a box

Choose **Box**. Turn **Static object** on for a fixed obstacle, or off for a body that can fall. Move onto terrain until the preview appears, then click and release left mouse to place it.

5. Add a ball

Choose **Ball** and turn **Static object** off. Move the preview near the box. Before placing, use the wheel to raise it above the terrain, then click and release. Page 3 explains sizing and rotation.

6. Save the scene

Click **Save scene** in **Editor > Objects**. This writes the authored level under **SkullbonezData/scenes/**. Check the save feedback before changing scenes or closing the app. **F2** makes a separate scene shot; it does not save the active level.

7. Run your experiment

Click **Edit scene** to leave editing. If paused, click **Play** on the bottom transport. Watch the ball fall and collide. Save the starting arrangement before running if you want to return to it.

8. Reopen and try again

Open **Tools > Scene**, use the scene dropdown's filter to find your level, and select it. Make another edit, save, and run again. A saved scene and a saved replay recording are different things (page 4).

Good to know

Creating a scene does not replace saving it. Use **Save scene** for your authored level. Tutorial practice scenes cannot be overwritten; create your own level to keep changes.



Place, select, transform

The pointer acts on the viewport. A click over a panel operates that panel instead.

Mode	Use it for
Choose a type in the object list	Preview and insert a new object. Placement enters Edit mode automatically.
Select [Q]	Select an existing object, then choose Move, Rotate or Scale. A gizmo is the visible set of transform handles.

Placing objects

Gesture or control	Result
Left click and release	Place the current preview. Placement commits on release.
Hold left mouse + drag	Resize before placing. Boxes use drag for footprint dimensions and wheel for height during the drag; round objects scale uniformly.
Wheel before clicking	Raise or lower the preview above its terrain placement.
Ctrl + wheel	Rotate the preview around the vertical axis in 15-degree steps.
Static object / Terrain align	Choose fixed or dynamic physics / align placement to the terrain slope.

Editing existing objects

Click **Select [Q]**, then click an object. Keep **Modify velocity** off when editing its shape or position. Choose the tool using its button or the bracketed shortcut while right mouse is released.

Tool or gesture	Result
Move [W]	Drag an arrow along one axis, or a square handle in a plane.
Rotate [E]	Drag a ring to rotate around its axis.
Scale [R]	Drag the centre for uniform size, or an axis for supported shape dimensions.
Space: World / Local	Choose fixed world axes or axes aligned to the object.
Move / Angle / Scale snap	Toggle snapping and choose each increment with its value control.
Esc during a drag	Cancel the edit and restore the starting transform.
Delete / Ctrl+Z / Ctrl+Y	Delete / undo / redo. Ctrl+Shift+Z also redoes.

Camera: hold right mouse to look and use W/A/S/D to move. Release it to return to the selected tool. Middle-drag pans; Alt-left-drag orbits and Alt-right-drag dollies. **Frame selected [F]** fits the selection in view.

Good to know

Use undo before leaving Edit mode; leaving it clears the edit history. Save your starting scene. A grey axis can mean the selected shape cannot represent that change.



Physics, prediction & replay

Start simple: save a scene, run it, inspect what happens, then reload your saved setup.

Control the simulation

Leave **Edit scene**, then use **Play/Pause** on the bottom transport. Single-step advances a paused simulation once. Physics controls provide body properties and diagnostic overlays: static bodies stay fixed; dynamic bodies can move or sleep at rest.

1. Predict and inspect

Open **Replay** and click the bold **Predict** control, then click a body in the viewport. Wait for its future path and **Causes**. Click a cause to inspect a collision; **Exit inspect** or **Esc** leaves inspection without launching the paused experiment.

2. Compare a different future

Modify velocity sits directly below Predict. With a predicted object selected, enable it, drag a velocity arrow and release. Compare **Original** and **Modified** by scrubbing the timeline. Choose **Accept Original** or **Accept Modified**, turn Predict off, then press **Play** to launch it.

3. Rewind or branch

Move to the bottom edge to reveal the timeline and drag its thumb to view recorded history. **Branch from here [Enter]**, at the bottom of Replay below Load recording, resumes from the selected recorded point. Save a recording first if you want a separate copy of the earlier run.

Understand the different saves

Action	What you keep
Editor > Objects > Save scene	The editable authored level under SkullbonezData/scenes/.
F2	A separate numbered scene shot; it does not save the active level.
F3	A viewport screenshot.
Replay > Save / Load recording	Recorded simulation data, separate from the editable scene. High detail prediction is just above Save recording.

Good to know

Reset rebuilds the scene; **Reset Defaults** reloads authored defaults and discards live edits. Save anything you want to keep before resetting or loading another level.

Explore further

The tutorial finale loads **solar_system.scene.json** from Tools > Scene. Set **Prediction horizon** to at least **40s**, select **Earth**, click **Predict**, wait for its lines, then press **Play**. The tutorial leaves that scene running.

Solver Lab opens separately supplied comparisons from Tools > Scene. The compact public ZIP omits the large built-in recordings. In Solver Lab, Space plays/pauses and left/right arrows step.



Keyboard & mouse

Shortcuts are context-sensitive. Finish typing in a filter or close a popup before using world controls. Tool shortcuts apply with right mouse released; vehicle and launcher modes have their own controls.

Key / gesture	Action and context
Right mouse + move mouse	Viewport camera look; pan in an orthographic view.
Right mouse + W/A/S/D; Q/E	Fly forward/left/back/right; down/up.
Shift / wheel while looking	Temporary faster travel / adjust flight speed.
Middle-drag / idle wheel	Pan / zoom.
Alt-left / Alt-right / Alt-middle drag	Orbit / dolly / pan around the current focus.
Q / W / E / R	Select / Move / Rotate / Scale in editing and free inspection.
F	Frame the selection in editing and free inspection.
Backtick (`)	Enter or leave Edit mode (US keyboard key below Esc).
Double-click / Esc	Attach to an object / detach. Esc dismisses or cancels an active interaction first.
T	Enter or leave Drive when attached to a car.
W/A/S/D / Space in Drive	Accelerate, steer, reverse / brake.
Ctrl+Z / Ctrl+Y or Ctrl+Shift+Z	Undo / redo an editor edit.
Delete / Esc during a transform	Delete the editable selection / cancel the drag.
Ctrl + wheel in placement	Rotate the preview by 15 degrees per wheel step.
P	Toggle paused prediction inspection. Use visible Play to run afterwards.
F1 / Enter when attached	Cycle attached-camera submode / toggle camera pin.
Enter in replay	Branch from the selected recorded point when available.
F2 / F3	Save a separate scene shot / screenshot.
0 / F7	Show/hide UI / switch Split Future and legacy lighting.
N / M	Toggle launcher / cycle its fire mode while active.
Space / left-right in Solver Lab	Play/pause / step backward-forward.
F5 / F6	Performance histogram / memory overlay.
V / C / G / O	Collision visuals / physics overlays / broadphase bounds / terrain contact probe.
[/] / 6	Previous/next physics pipeline stage / transparent debug display.
1 / 2 / 3; 4 / 5	Freeze water / reflection mode / flat water; show/hide terrain / water.

These are the main first-session controls, not every developer shortcut. **Tools > Keys**, hover help and the active mode provide more context. Use the visible **Reset** button: **R** is Scale in the usual editor view.



When something feels stuck

Most first-session surprises come from an active mode, a paused scene, or a pointer over a panel.

What you see	What to check
Objects will not move	Leave Edit scene, turn Predict off and press Play. Check the body is dynamic rather than Static.
Clicks keep adding objects	Choose Select [Q] before selecting an existing object.
The cursor disappears	Release right mouse. A placement preview can replace the normal cursor over the world.
No placement preview	Choose an object type and move over terrain inside the viewport, clear of panels.
Shortcuts do nothing	Finish editing the text field or close the popup. Check the mode and release right mouse for tool shortcuts.
Controls are missing	Scroll inside the panel, open its dock or enlarge the window. Press 0 if the UI is hidden.
My scene is missing after restart	New scene alone does not save it. Use Editor > Objects > Save scene and check the save feedback.
Files do not save or assets are missing	Extract the whole ZIP to a writable folder and use Launch SkullbonezCore.cmd. Do not run from inside the ZIP.
A tutorial step is stuck	Use Retry section for a fresh chapter setup. Resume restores a chapter, not a previous session's temporary recording.

Terrain and water

In Edit mode, open **Terrain**, enable **Terrain brush**, then hold **left mouse** to paint. Use **Brush: Raise ground / Lower ground** to choose the direction and **Brush radius** to set its size. Right-drag moves the camera. Use **Water level** and **Show water** to make a shoreline. Save your authored scene afterwards.

A useful first practice session

Create a new scene, place one static box and one dynamic ball, save, leave editing and watch. Reload, move the ball higher, save again and use **Predict** to explore the result. Or follow **Tutorial** from its camera lesson through the Solar System finale.

About this guide

A practical guide to **SkullbonezCore 2026.09.23.2**, covering the editor, camera navigation, scene saving, Replay and tutorial controls a new user needs first.

[Downloads and feedback](#) - include your build version, what you did and what happened when reporting a problem. The online **LLM Level-Authoring Guide** is linked from the repository README.